

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1818

Little Spadra Creek 138 kV
July 19, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1818. The Interconnection Customer proposes a 275.8 MW solar photovoltaic facility interconnecting at the Little Spadra Creek 138 kV substation in the Maryland area. The request was received on July 18, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 429 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,130,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Little Spadra Creek-Posco 138 KV	Campus Drive-Bladensburg 138 KV	128.0	222	27.8
Posco-Rush 138 KV line	Campus Drive Substation-Bladensburg 138 KV	122.9	215	27.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Little Spadra Crk-Posco 138 KV	Rebuild line with higher-capacity conductor	\$1,446,000	72% (\$1,049,000)
Posco-Rush 138 KV	Rebuild line with higher-capacity conductor	\$1,491,000	72% (\$1,081,000)
	Total allocated to project		\$2,130,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.